

Dermatologic Conditions in Kidney Disease

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CHAPTER OUTLINE

SKIN MANIFESTATIONS SECONDARY TO KIDNEY DYSFUNCTION, 1932
 NAIL CHANGES ASSOCIATED WITH KIDNEY DISEASE, 1940
 SELECTED CONDITIONS WITH SKIN AND RENAL INVOLVEMENT, 1940

GENODERMATOSES, 1942
 DERMATOLOGIC CONDITIONS THAT MAY LATER INVOLVE THE KIDNEYS, 1943

KEY POINTS

- Skin manifestations of end-stage renal disease are diverse, with pruritus being very common.
- Calciphylaxis may lead to high morbidity and mortality. Skin lesions range morphologically from a broken-up, lacelike pattern of pink to purple erythema early on (reticulated vascular pattern) to ulcers, with black eschars in later stages.
- Pseudoporphyria and porphyria cutanea tarda may have very similar clinical presentations. Normal levels of plasma uroporphyrin and red blood cell protoporphyrin are typical of the former.

Many patients with end-stage renal disease (ESRD) have associated skin manifestations.^{1,2} In addition, skin findings may specifically direct an astute clinician to check for concomitant renal dysfunction. This chapter summarizes different dermatologic conditions that may be seen in patients with kidney disorders. Because of the large number of diseases falling into this category (Boxes 58.1 to 58.4 and Table 58.1), emphasis is placed on the more commonly encountered entities.

SKIN MANIFESTATIONS SECONDARY TO KIDNEY DYSFUNCTION

See Box 58.1.

SIGNS AND SYMPTOMS

PRURITUS

Pruritus is more common in patients with ESRD disease than in those with acute kidney injury (AKI). Up to 90% of patients undergoing hemodialysis may experience pruritus, and patients undergoing hemodialysis are more commonly affected than patients undergoing peritoneal dialysis.² In one of the largest trials (the Dialysis Outcomes and Practice Patterns Study [DOPPS]), pruritus was noted in 42% of hemodialysis patients. Sleep and mood are negatively affected.³ Patients

with severe pruritus have a poor outcome compared with patients undergoing hemodialysis without severe pruritus.⁴

Patients complaining of pruritus may or may not have skin changes. Pruritus can be defined as at least three episodes of itch in a 2-week period that cause difficulty for the patient or as itch that occurs over a 6-month period in a regular pattern.³ Pruritus can be localized or generalized.⁵ When present, skin manifestations are secondary, with excoriations being the main finding (Fig. 58.1). Lichenified skin, prurigo nodularis, and koebnerization (the appearance of skin lesions on lines of trauma) may also be seen. Pruritus tends to be prolonged, frequent, and intense.³ Exacerbating factors include heat, nighttime dry skin, and sweat.³

The cause of pruritus in renal failure is unclear and may be multifactorial. Risk factors include male gender and high levels of urea, β_2 -microglobulin, calcium, and phosphate.^{4,6} Patients treated with angiotensin-converting enzyme inhibitors are more likely to have pruritus than those receiving furosemide.³ Decreasing urine output, secondary hyperparathyroidism, abnormal levels of magnesium and aluminum, increased levels of histamine and vitamin A, increased numbers of mast cells,⁷ xerosis,⁸ and iron deficiency anemia are other proposed factors.^{9,10}

Pruritus is transmitted through C fibers in the skin.¹¹ C group fibers are unmyelinated and have a small diameter and low conduction velocity. They represent one of three

Box 58.1 Skin Manifestations Secondary to Renal Disease**Nonspecific**

Pruritus
 Xerosis
 Acquired ichthyosis
 Pigmentary alteration

- Pallor (secondary to anemia)
- Hyperpigmentation
- Dyspigmentation (yellow tint)

 Infections (fungal, bacterial, viral)
 Purpura

Somewhat Specific

Acquired perforating dermatosis
 Calciphylaxis
 Metastatic calcification
 Blistering disorders

- Porphyrria cutanea tarda
- Pseudoporphyria

 Eruptive xanthomas
 Pseudo-Kaposi sarcoma

Specific

Nephrogenic systemic fibrosis
 Dialysis-associated steal syndrome
 Metastatic renal cell carcinoma
 Dialysis-related amyloidosis
 Arteriovenous shunt dermatitis
 Uremic frost



Fig. 58.1 Pruritus. Shown is a patient with linear excoriations. (Courtesy Dr. Oscar Colegio.)

classes of nerve fibers in the peripheral nervous system and carry sensory information. They are afferent fibers, sending input signals from the periphery to the central nervous system (CNS). Known stimulants of C fibers include cytokines, histamine, serotonin, prostaglandins, neuropeptides,⁵ and enzymes.¹² Because cytokines can stimulate nerve fibers,¹² one major theory regarding the pathogenesis of pruritus in renal failure is that it involves systemic inflammation.¹³ Markers of inflammation, such as C-reactive protein and interleukin-6, show elevated levels in pruritus associated with renal disease.^{13,14} An imbalance of inflammatory proteins may somehow lead to pruritus. This theory is supported by the fact that ultraviolet light treatments, which can decrease the levels of inflammation markers,¹⁵ often ameliorate pruritus.

Another major theory focuses on the opioid system. Opioids can stimulate C fibers.¹² Central μ -opioid receptor stimulation in mice leads to scratching, and this can be prevented by central κ -opioid receptor stimulation.¹⁶ Successful treatment of patients with opioid antagonists also supports the involvement of the opioid system in pruritus.^{17–19} Other potentially pathogenic factors include xerosis, elevated histamine, excess uremic toxins, peripheral neuropathy, and hyperparathyroidism.²

Treatment of pruritus must be tailored to the individual patient. In general, antihistamines are not effective.²⁰ Any condition that can exacerbate pruritus, such as xerosis, should be treated.²¹ Symptoms may be alleviated by optimization of hemodialysis¹ and administration of erythropoietin.²² For more localized areas of pruritus, topical corticosteroids, pramoxine (also known as pramocaine),²³ and capsaicin^{5,24} may be used. In addition, twice-weekly ultraviolet light phototherapy is effective.²⁵ Oral treatments include gamma-linoleic acid,²⁶ naltrexone,^{17,18} cholestyramine,²⁷ gabapentin,^{28,29} pregabalin,³⁰ thalidomide,³¹ and activated charcoal.³² κ -Opioid receptor agonists such as nalfurafine¹⁹ and nalbuphine³³ may be helpful. Kidney transplantation may effectively “cure” patients,³⁴ as may parathyroidectomy.³⁵

A therapeutic ladder has been proposed, based on efficacy and safety considerations.³⁴ The first rung of the ladder consists of emollients and capsaicin. If pruritus is not relieved, ultraviolet light treatments can be added. Phototherapy with ultraviolet B (UVB; 280–320 nm) is effective in reducing pruritus in dialysis patients and may act by suppressing histamine release or vitamin A levels in the epidermis, which are known pruritogenic substrates. Importantly, the risk of skin cancer should be balanced with decision making, especially in patients who are on immunosuppressive therapy. UVB radiation is absorbed in the epidermis and superficial dermis by molecules called “chromophores,” which include DNA, urocanic acid, keratin, and melanin. UVB radiation induces the release of a variety of proinflammatory and immunosuppressive cytokines from keratinocytes and T cells. If light treatments fail or are not feasible for the patient, oral gabapentin or intravenous nalfurafine may be considered.

XEROSIS

Xerosis, or dry skin, is quite common in the general population. The skin appears dry, rough, or shiny. It may be scaly or fissured, with a cracked appearance. Xerotic skin may or may not be pruritic.²¹ If it is pruritic, there may be excoriations. Skin involvement may be diffuse but is often most prominent over the extensor surfaces of the lower extremities (Fig. 58.2).¹

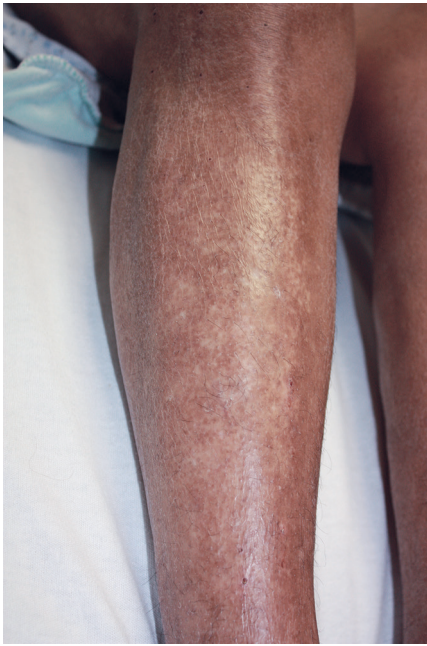


Fig. 58.2 Xerosis and dyspigmentation. (Courtesy Dr. Marcus McFerren.)

The cause of xerosis is unknown. Decreased stratum corneum hydration²¹ and abnormal eccrine gland function³⁶ are two proposed mechanisms. The mainstay of treatment is hydration of the skin. For pruritic cases, direct treatment of the pruritus may be helpful.

ACQUIRED ICHTHYOSIS

Ichthyosis is related to xerosis but is more than just dry skin because the skin develops patterned scale. Histopathologic features include hyperkeratosis and occasionally epidermal hypogranulosis. The pathogenesis of acquired ichthyosis in renal failure is unknown. Treatment consists of hydration of the skin.

PIGMENTARY ALTERATION

Of patients undergoing dialysis, 70% or more may have pigmentary changes of the skin.^{2,37} The most common alteration is the development of a yellowish tint. This is more common among patients undergoing hemodialysis than those on peritoneal dialysis.² Hyperpigmentation may be secondary to increased melanin production as a result of elevated levels of β -melanocyte-stimulating hormone.³⁸ Pallor of the skin secondary to anemia has also been described.¹

MANIFESTATIONS SOMEWHAT SPECIFIC TO KIDNEY DISEASE

ACQUIRED PERFORATING DERMATOSIS

An umbrella term for perforating disorders in adults with kidney dysfunction and/or diabetes mellitus is “acquired perforating dermatosis.” This term is often used in place of older nomenclature (e.g., perforating folliculitis, Kyrle disease, reactive perforating collagenosis). Some authors consider this disorder to be a variant of prurigo nodularis.³⁹ Pruritus



Fig. 58.3 Acquired perforating dermatosis. (Courtesy Yale Residents' Collection.)

is generally severe.⁴⁰ Lesions are distributed predominantly on the legs and arms, although the trunk and head may also be involved. Individual lesions are crateriform, umbilicated, or centrally hyperkeratotic papules and nodules (Fig. 58.3). They may develop in crops and commonly resolve with scarring after 6 to 8 weeks. Histopathologic evidence of extrusion of dermal material through an epidermal channel is necessary for the diagnosis (Fig. 58.4).

The pathogenesis of this disorder is unknown. Proposed factors include increased fibronectin,⁴¹ pruritus and scratching,⁴⁰ epidermal dysmaturation, and dermal deposition of substances not excreted in renal failure.^{42,43} Patients with diabetes and renal failure are more likely to manifest acquired perforating dermatosis than patients without diabetes.^{1,40} Other associations include hepatitis and hypothyroidism.⁴³ Patients who are otherwise healthy may also develop acquired perforating dermatosis.⁴³ Treatment of this disorder is difficult, with some relief afforded by topical steroids, keratolytics, lubrication, and topical or oral retinoids.⁴² Narrow-band UVB light may also be helpful.⁴⁴ Patients with severe and generalized pruritus may benefit from treatments specifically directed at pruritus.

CALCIPHYLAXIS (CALCIFIC UREMIC ARTERIOLOPATHY)

Calciphylaxis is most often seen in patients with ESRD who are on hemodialysis, generally with associated hyperparathyroidism or an elevated calcium phosphate product. Although rare, calcific uremic arteriolopathy (CUA) can also present in chronic kidney disease (CKD). An incidence of approximately 4% of patients/year at one dialysis center has been reported.⁴⁵ Morbidity and mortality are high, especially once ulceration develops.⁴⁵ Involvement of the trunk and buttocks is associated with high mortality.⁴⁶ Other risk factors include female gender,⁴⁵ diabetes,⁴⁵ obesity, hypoalbuminemia,⁴⁷ malignancy, administration of systemic steroids, iron usage, warfarin use,⁴⁷ chemotherapy, systemic inflammation, hepatic cirrhosis, rapid weight loss, protein C or S deficiency, and infection.⁴⁸

The precise mechanism of calciphylaxis is unclear. The classic model, as described by Selye in rats, hypothesized a predisposing factor, with a secondary inciting factor.⁴⁹ This model has been criticized as not being fully reflective of calciphylaxis in humans, because Selye rats displayed only

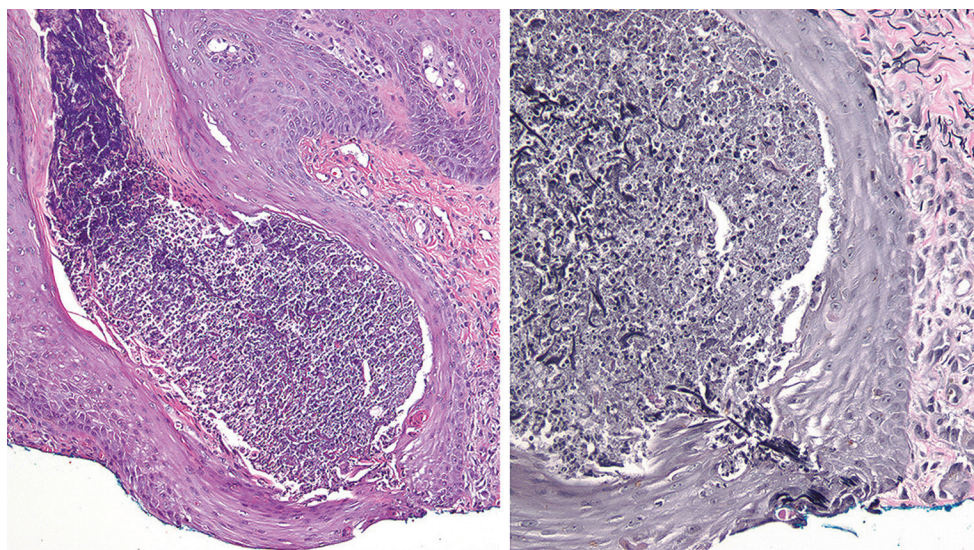


Fig. 58.4 Acquired perforating dermatosis. There is an epidermal channel with extrusion of dermal elastic material and inflammatory cells. (Left, Hematoxylin and eosin stain, $\times 40$; right, elastic van Gieson stain, $\times 100$.)



Fig. 58.5 Calciphylaxis. Shown is an early lesion with a retiform purpuric appearance.



Fig. 58.6 Calciphylaxis. Shown is an ulcerated lesion with black eschar. (Courtesy Yale Residents' Collection.)

soft tissue calcification and not skin necrosis.^{50,51} Although prior literature focused on calcium, phosphate, and inhibitors and inducers of vascular calcification,⁵¹ hypercoagulability may play a more important role.⁵²

Clinically, calciphylaxis usually affects the fatty areas of the thighs, abdomen, and buttocks symmetrically. Lesions may also affect the lower extremities.⁴⁵ Early lesions may be pink-violaceous, broken-up circles, firm plaques, solid purple-pink or mottled, retiform purpura, erosions, or subcutaneous nodules (Fig. 58.5).⁵³ Occasionally, bullae may be seen. Early lesions may progress to painful ulcers with a black eschar (Fig. 58.6). Surrounding the ulcers, there may be skin mottling, with reticulate dyspigmentation. The skin may be tender. Histopathologic findings supportive of calciphylaxis include medial calcification of medium-sized vessels (Fig. 58.7), with intimal hyperplasia and thrombosis. Special stains, such as von Kossa or alizarin red, may be required to detect early calcification. Radiographic studies and tissue mammography may also show linear calcium deposits

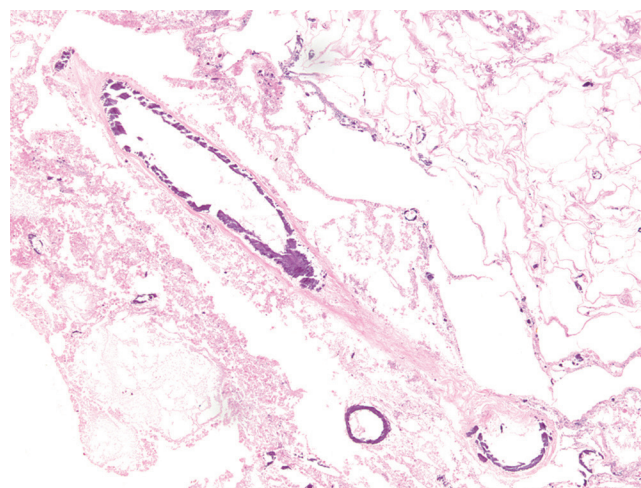


Fig. 58.7 Calciphylaxis. Shown are medium-sized vessels with extensive calcium deposition. (Hematoxylin and eosin stain, $\times 40$.)

in the skin. Laboratory evaluation should include renal function, evaluation of parameters related to bone mineralization (e.g., calcium, phosphate, parathyroid hormone), workup for thrombophilia, liver function, assessment for any infections based on the patient's presentation, and evaluation for autoimmune diseases, as clinically indicated. Deficiencies in inhibitors of vascular calcifications may play a role in CUA pathogenesis, especially fetuin-A and vitamin K-dependent matrix Gla protein. Polymorphisms in ecto-5'-nucleotidase CD73, vitamin D receptor, and fibroblast growth factor-23 (FGF-23) have been suggested to play a role in calciphylaxis development.⁵⁴

Supportive and preventive measures are the mainstay of treatment, including wound care, pain management, and modification of any risk factors. For nonulcerated lesions, systemic steroids may lead to rapid healing.⁴⁵ Ulcers should be kept clean, débrided if necessary, with the addition of hyperbaric oxygen therapy⁵⁵ and/or antibiotics, depending on clinical factors and the response. Potentially associated factors such as elevated calcium and phosphate levels can be addressed with intravenous sodium thiosulfate,⁵⁶ parathyroidectomy,^{45,50} oral cinacalcet and phosphate binders,⁵⁷ use of low-calcium dialysate and non-calcium-based phosphate binders,⁵⁸ and bisphosphonates.⁵⁹ If there is a hypercoagulable state, anticoagulation may be considered. If the patient is on warfarin, it is worthwhile to consider changing to another anticoagulant.^{51,52}

METASTATIC CALCIFICATION

An abnormal calcium phosphate product, with or without hyperparathyroidism, can lead to metastatic calcification of the skin. Clinically, lesions are hard, yellow to bluish papules and nodules, which usually affect periarticular areas and the fingertips.⁵⁸ A biopsy specimen from a nodule should reveal calcium deposits in the tissue. Importantly, not all calcification in the skin is secondary to metastatic calcification. Trauma may result in so-called dystrophic calcification. Some systemic diseases, such as dermatomyositis, may be associated with calcification of the skin. Calcification may also be idiopathic.

Treatment of metastatic calcification focuses on normalizing calcium and phosphate levels. If hyperparathyroidism is present, parathyroidectomy may be beneficial. Use of phosphate binders and reduction of dietary phosphate are important. Foods that should be avoided include milk and milk products, certain vegetables (e.g., broccoli, Brussels sprouts), oysters, salmon, beer, nuts, and wheat germ.⁵⁸ If particular lesions are symptomatic, surgical removal may be considered.

PORPHYRIA CUTANEA TARDA

Up to 18% of patients with renal failure may have porphyria cutanea tarda, with a deficient activity of the heme biosynthetic enzyme uroporphyrinogen decarboxylase (UROD) in the liver. These patients have elevated levels of urinary uroporphyrin (if not anuric) and fecal isocoproporphyrin. The pathogenesis of porphyria cutanea tarda in renal patients is unclear, yet reduced UROD activity is responsible for the accumulation of porphyrinogens that are auto-oxidized to photosensitizing porphyrins. These cause photodamage on exposure to light with a wavelength near 400 nm. Heterozygosity for a UROD mutation may predispose some patients, with



Fig. 58.8 Porphyria cutanea tarda. Shown are eroded blisters and crusts. (Courtesy Yale Residents' Collection.)

acquired functional UROD deficiency being superimposed. The serum aluminum level may also be a contributing factor.⁶⁰ Other risk factors for the development of porphyria cutanea tarda include hepatitis B and C, estrogen administration, excess alcohol intake, and human immunodeficiency virus (HIV) infection.

Cutaneous findings include noninflamed blisters, erosions, and crusts (Fig. 58.8), especially those involving the dorsal hands and forearms. Blisters may heal with scarring or milia. There may be notable hypertrichosis on the face. Hyperpigmentation and sclerodermoid plaques have also been described. Biopsy findings typically include a subepidermal cleft, with minimal inflammation. There may be festooning of the papillary dermis at the base of the cleft. Thickened vessel walls may be present. Direct immunofluorescent findings in porphyria cutanea tarda include granular to linear staining of immunoglobulin G (IgG) and C3 at the dermoepidermal junction and sometimes around vessels.

Treatment includes sun avoidance and sun protection. Concomitant exacerbating factors (e.g., alcohol, estrogens, iron, hepatitis) should be avoided or treated. Because iron overload can exacerbate the disease, small-volume phlebotomy may be helpful, although anemia needs to be carefully monitored. Desferoxamine, an iron chelator, is also sometimes used. If a patient is not anuric, chloroquine may be used cautiously.⁵⁸

PSEUDOPORPHYRIA

In pseudoporphyria, patients have a presentation similar to that in porphyria cutanea tarda, with noninflamed blisters on the extremities, especially the hands and forearms and other sun-exposed areas. Milia, scarring, and hyperpigmentation may result.⁶¹ Hypertrichosis is seen in a minority of patients.⁶¹ Plasma uroporphyrin and red blood cell protoporphyrin levels rarely are elevated.⁶⁰ The separation of pseudoporphyria from porphyria cutanea tarda is clearer when these levels are normal.⁶¹ Concomitant use of drugs such as tetracycline, furosemide, naproxen, amiodarone, and nalidixic acid may be involved in the development of pseudoporphyria.⁶¹ Histologic findings are similar to those of porphyria cutanea tarda.⁶¹ Although the pathogenesis of pseudoporphyria is unclear, elevated aluminum levels were found in a small study.⁶⁰

ERUPTIVE XANTHOMAS

Eruptive xanthomas are yellow-orange smooth papules and plaques (Fig. 58.9) that rapidly appear on the buttocks and proximal extremities. Histopathologic examination of a lesion reveals extracellular lipid and foamy macrophages (Fig. 58.10). These lesions are associated with hyperlipidemia, either familial or due to other causes, such as hypothyroidism and nephrotic syndrome. In nephrotic syndrome, the xanthomas are likely secondary to various lipid abnormalities, including elevated levels of cholesterol and triglycerides, elevated lipoprotein synthesis, diminished level of lipoprotein lipase, and elevated apolipoprotein B-100 level, with diminished hepatic uptake of lipoproteins.⁶² Generally, eruptive xanthomas resolve spontaneously once lipid levels are normalized.

PSEUDO-KAPOSI SARCOMA

Very rarely, patients may develop a vascular proliferation that mimics Kaposi sarcoma near or over an arteriovenous shunt.⁶³

IODODERMA

Often, in the setting of renal insufficiency or failure, patients who receive iodide in the form of intravenous contrast may

develop translucent papulonodular or vegetative lesions of iododerma (Fig. 58.11).⁶⁴ Histopathologically, there is marked epidermal hyperplasia, with intraepidermal pustules of neutrophils and sometimes eosinophils (Fig. 58.12). Clinical and histopathologic exclusion of infectious causes is required.

MANIFESTATIONS SPECIFIC TO KIDNEY DISEASE

See Box 58.1.

NEPHROGENIC SYSTEMIC FIBROSIS

The first case of nephrogenic systemic fibrosis (NSF), originally described under various terms, including “scleromyxedema-like changes in renal failure” and “nephrogenic fibrosing



Fig. 58.9 Eruptive xanthomas. Shown are small, yellow-orange papules over the knees. (Courtesy Yale Residents' Collection.)



Fig. 58.11 Iododerma. Shown are translucent papules and crusted plaques on the elbow. (Courtesy Dr. Oscar Colegio and Dr. Christine Warren.)

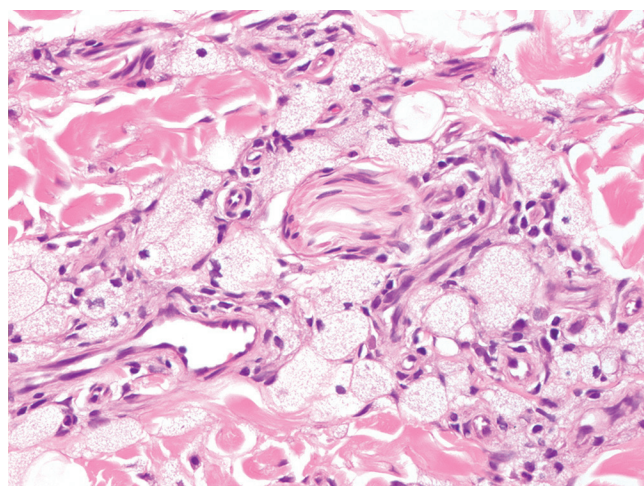


Fig. 58.10 Eruptive xanthoma. Shown are lipid-containing foamy macrophages within the dermis. (Hematoxylin and eosin stain, $\times 40$.)

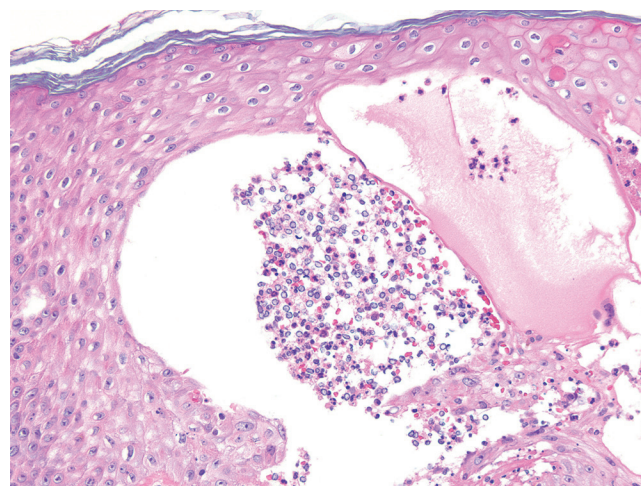


Fig. 58.12 Iododerma. Shown is epidermal hyperplasia with intraepidermal pustules. (Hematoxylin and eosin stain, $\times 200$.)



Fig. 58.13 Nephrogenic systemic fibrosis. The lower leg has indurated, bound-down skin and contractures of the toes. (From Cowper SE, Rabach M, Girardi M. Clinical and histological findings in nephrogenic systemic fibrosis. *Eur J Radiol.* 2008;66[2]:191–199.)



Fig. 58.14 Nephrogenic systemic fibrosis. The legs have the cobblestone appearance of indurated skin. (From Girardi M. Nephrogenic systemic fibrosis: a dermatologist's perspective, *J Am Coll Radiol.* 2008;5[1]:40–44.)



Fig. 58.15 Nephrogenic systemic fibrosis. Shown is a joint contracture. (From Girardi M. Nephrogenic systemic fibrosis: a dermatologist's perspective, *J Am Coll Radiol.* 2008;5[1]:40–44.)

dermopathy,⁶⁵ was identified in 1997. Although the cause of NSF was long debated, exposure to gadolinium-based magnetic resonance imaging (MRI) contrast agents in patients with abnormal renal function is now considered the precipitating factor.^{66,67} The U.S. Food and Drug Administration (FDA) recommends caution when using gadolinium in patients with an estimated glomerular filtration rate (eGFR) less than 30 mL/min/1.73 m², undergoing dialysis, or with AKI. The latter is especially important given that it is difficult to predict eGFR in a non-steady-state setting. Gadolinium, an element highly toxic to tissues in its unbound state, is present in a chelated form in contrast agents such as gadodiamide (Omniscan; GE Healthcare, Piscataway, NJ) and gadopentetate dimeglumine (Magnevist; Bayer HealthCare, Whippany, NJ). Gadolinium is a nonradioactive paramagnetic element in the lanthanide series. Impaired excretion of gadolinium-based contrast may allow more time for gadolinium atoms to dissociate from their proprietary ligand molecule. Unbound gadolinium is theorized to bind to other available anions (chiefly phosphates) and deposit peripherally, perhaps inducing long-standing effects on local tissue fibroblasts and/or circulating matrix stem cells termed “circulating fibrocytes.” Gadolinium has been detected in tissues of patients with NSF and in in vivo animal studies, findings that corroborate the postulated effects of prolonged gadolinium exposure.⁶⁸

Clinically, patients with NSF have bound-down, indurated skin on the extremities (Fig. 58.13), although truncal involvement has also been reported. The skin may have a cobblestone appearance (Fig. 58.14). Early in NSF, patients may have edema and erythema that mimics cellulitis. Joint contractures (Fig. 58.15; see Fig. 58.13) and yellow scleral plaques

(Fig. 58.16) are common. Histopathologically, increased dermal and/or subcutaneous fibroblast-like cells are seen (Fig. 58.17). A deep biopsy may be needed because the fibrotic lesions can extend into the subcutaneous tissue. The cells stain with procollagen I and CD34 by immunohistochemical methods. It is postulated that these cells are circulating fibrocytes.⁶⁹ Mucin deposition may also be observed. The prevalence of systemic involvement is unknown, but a number of different organ system manifestations have been described (e.g., heart, muscle, lung). Patients with systemic disease may have marked elevations in their erythrocyte sedimentation rate and serum C-reactive protein level. The clinical and histopathologic differentials are broad and overlap with some more common entities, such as lipodermatosclerosis and morphea. Clinicopathologic correlation is essential to reach an accurate diagnosis.⁷⁰



Fig. 58.16 Nephrogenic systemic fibrosis. Shown is a scleral plaque. (From Cowper SE, Rabach M, Girardi M. Clinical and histological findings in nephrogenic systemic fibrosis. *Eur J Radiol.* 2008;66[2]:191–199.)

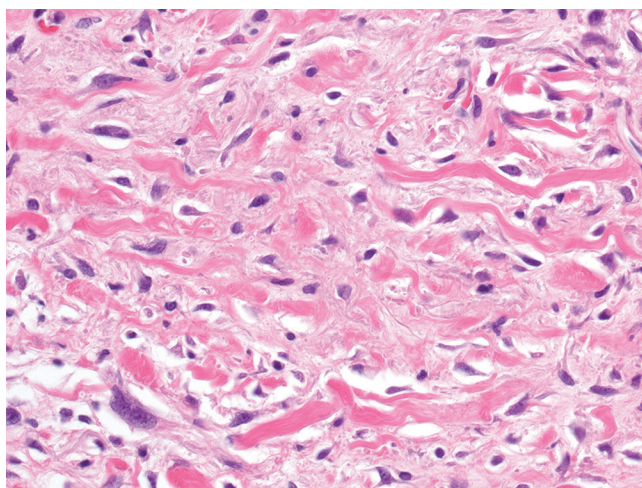


Fig. 58.17 Nephrogenic systemic fibrosis. There are increased numbers of fibroblast-like cells, preserved elastic tissue, and increased space between collagen bundles, suggestive of mucin deposition. (Hematoxylin and eosin stain, $\times 200$.)

Prevention of NSF by avoiding the administration of gadolinium-containing agents to patients with CKD or ESKD is key because there is no reliable treatment for the disease. There is evidence that a dose-response relationship exists. The risk in individuals with eGFR of 15 to 59 mL/min/1.73 m² remains unclear, but cases have been reported. Changes in the labeling of gadolinium-containing contrast agents and careful screening of patients for underlying kidney disease have been successful preventive measures. In patients who develop the disorder, renal transplantation is sometimes helpful in halting progression and occasionally reverses the disease.⁷¹ Extracorporeal photopheresis may also be useful. Physical therapy may ameliorate some symptoms for patients. Reports of imatinib treatment are encouraging.⁷²



Fig. 58.18 Dialysis-associated steal syndrome. Shown is an ulcer with overlying crust, with surrounding erythema, distal pallor, and nail atrophy. (From Kravetz JD, Heald P. Bilateral dialysis-associated steal syndrome. *J Am Acad Dermatol.* 2008;58:888–891,2008.)

DIALYSIS-ASSOCIATED STEAL SYNDROME

Dialysis-associated steal syndrome is a rare complication of fistula construction.⁷³ This syndrome usually involves the brachial area in patients with diabetes. There may be associated pallor or a reticulated pink to blue discoloration of the skin surrounding necrosis, ulceration, or gangrene (Fig. 58.18). Altered hemodynamics secondary to the fistula result in decreased distal perfusion. Associated risk factors are diabetes, vascular stenosis, neuropathic disease, and calcifying sclerosis. Effective treatment involves fistula ligation and/or banding.

METASTATIC RENAL CELL CARCINOMA

Although approximately 4.6% of cutaneous metastases originate from the kidney, only 3% to 4% of renal cell carcinomas metastasize to the skin.⁷⁴ Metastases may occasionally precede the diagnosis of the primary tumor, but often they are a late manifestation.⁷⁵ The prognosis is poor, with a 5-year survival of less than 5%. Common cutaneous sites for metastatic renal cell carcinoma are the trunk and scalp.⁷⁵ Nodules may be flesh-colored, violaceous, or pink-red. Histopathologic examination often shows a tumor with clear cells and prominent hemorrhage. The latter is a feature of the highly vascular nature of von Hippel-Lindau (VHL)-deficient clear cell renal cell carcinoma.

DIALYSIS-RELATED AMYLOIDOSIS

Dialysis-related amyloidosis is secondary to β_2 -microglobulin deposition. Unlike AL amyloid, skin involvement is unusual, and the more common presentations involve carpal tunnel syndrome and destructive arthropathy. However, there have been rare reports of immobile dermal nodules secondary to β_2 -microglobulin deposition, most often affecting the buttocks.^{76,77}

ARTERIOVENOUS SHUNT DERMATITIS

In one study, 7 of 88 patients on long-term hemodialysis developed an irritant contact dermatitis over their shunts.⁷⁸

None of the patients had an allergic component to the rash. Substituting normal saline for the cleansers (soaps, disinfectants, and alcohol) used to clean the skin before hemodialysis was helpful in alleviating the skin rash. Use of a mild topical steroid was also helpful.

NAIL CHANGES ASSOCIATED WITH KIDNEY DISEASE

See [Box 58.2](#).

LINDSAY (HALF-AND-HALF) NAIL

Up to 40% of patients with renal dysfunction have half-and-half nails ([Fig. 58.19](#)).² The condition may resolve spontaneously and is likely secondary to melanin deposition in the nail bed and plate. Fingernails are affected more commonly than toenails. The nails have a white to normal proximal half and a red-brown distal half.

SELECTED CONDITIONS WITH SKIN AND RENAL INVOLVEMENT

See [Box 58.3](#).

Box 58.2 Nail Changes Associated With Kidney Disease

Lindsay (half-and-half) nail
Triangular lunulae (nail patella syndrome)
Splinter hemorrhages
Onychomycosis
Koilonychia
Onycholysis
Mees lines
Muehrcke lines
Beau lines



Fig. 58.19 Lindsay (half-and-half) nail. (From Butler DF. Pruritus. In Schwarzenberger K, Werchniak A, Ko C, eds. *General Dermatology*. Philadelphia: Saunders; 2009:17–22.)

LUPUS ERYTHEMATOSUS

The skin manifestations of lupus erythematosus are varied. Acute skin changes include the classic malar rash. Annular plaques and papules with scale and erythema are typical of subcutaneous lupus erythematosus. These lesions generally resolve without scarring. In discoid lupus erythematosus, coin-shaped to oval plaques heal with dyspigmentation and scarring. There is often adherent scale within the lesions that, when lifted off, has a “carpet tack” appearance on the underside. Such discoid lesions in the conchal bowl of the ear are characteristic of lupus erythematosus ([Fig. 58.20](#)). Neonates with lupus erythematosus may occasionally have annular, erythematous plaques in sun-exposed areas. In addition to these skin lesions, patients may complain of photosensitivity, oral ulcers, or alopecia.

The different skin manifestations of lupus look similar histopathologically. There are vacuolar changes at the dermoepidermal junction, with a variable infiltrate of lymphocytes. There is often a superficial and deep perivascular and periadnexal lymphocytic infiltrate as well. Mucin may be increased. Direct immunofluorescence testing may show a discontinuous linear band of IgG, C3, IgA, and/or IgM at the dermoepidermal junction.

The pathogenesis of lupus erythematosus involves auto-antibodies that can form immune complexes that deposit

Box 58.3 Selected Conditions With Concurrent Skin and Renal Involvement

More Common

Lupus erythematosus
Leukocytoclastic vasculitis
Henoch–Schönlein purpura
Mixed cryoglobulinemia
Diabetes mellitus
Systemic vasculitis

Less Common

Nail–patella syndrome
Hemolytic-uremic syndrome
Toxic shock syndrome
Mixed connective tissue disease
Dermatomyositis
Rheumatoid arthritis
Sjögren syndrome
Dermatitis herpetiformis
Sarcoidosis
Systemic sclerosis
Ulcerative colitis
Amyloidosis
Toxic epidermolysis
Hypothyroidism
Graves disease
Fabry disease
Neurofibromatosis
Hurler syndrome
Castleman disease
Infectious endocarditis
Staphylococcal scalded skin syndrome (in adults)



Fig. 58.20 Discoid lupus erythematosus. (Courtesy Yale Residents' Collection.)

in end-organs, causing damage.⁷⁹ Animal models of lupus have suggested that inhibition of oxidative stress and production of nitric oxide via inactivation of the nuclear factor- κ B pathway may be important in modulating disease.⁸⁰

Skin manifestations of lupus erythematosus are treated with steroids (oral or topical), antimalarials, and/or other immunosuppressants.

LEUKOCYTOCLASTIC VASCULITIS

Leukocytoclastic vasculitis, clinically known as “palpable purpura,” may be seen in a variety of clinical situations. Characteristically, the lesions are found on the lower extremities as nonblanching to partially blanching papules and plaques (Fig. 58.21). Histopathologically, there is leukocytoclasia, fibrin thrombi within vessels, extravasated erythrocytes, and swollen endothelial cells. If direct immunofluorescence testing is performed on the skin, deposits of IgG, IgM, and/or C3 may be seen around the vessels.

Associated diseases include Henoch–Schönlein purpura, infections (e.g., streptococcal, mycoplasmal, viral), systemic vasculitis (granulomatosis with polyangiitis [formally designated as Wegener granulomatosis], eosinophilic granulomatosis with polyangiitis [formally designated as Churg–Strauss syndrome], polyarteritis nodosa), inflammatory bowel disease, and malignancy.⁸¹ Lesions may also be drug-induced. In Henoch–Schönlein purpura, the skin lesions are often seen in association with gastrointestinal pain, joint pain, and renal involvement. Mixed cryoglobulinemia is another disorder that can present with palpable purpura in the skin, hypocomplementemia, and a membranoproliferative glomerulonephritis.⁸² Other skin manifestations include ulcers and



Fig. 58.21 Leukocytoclastic vasculitis. (Courtesy Yale Residents' Collection.)



Fig. 58.22 Henoch–Schönlein purpura. Nonblanchable macules and papules are distributed over the buttocks and lower extremities.

dyspigmentation. Many cases of mixed cryoglobulinemia are associated with hepatitis C infection; however, a variety of other infections, systemic diseases, or lymphoproliferative processes may also be associated.

HENOCH–SCHÖNLEIN PURPURA

Henoch–Schönlein purpura typically involves the skin (Fig. 58.22), joints, gastrointestinal tract, and kidneys.⁸³ Children

between the ages of 3 and 10 years are most commonly affected. In one series, 82% of affected children had joint pain, 63% had abdominal pain, 33% had gastrointestinal bleeding, and 40% had nephritis.⁸⁴ In a cohort of 250 adults, 61% had arthritis, 48% had gastrointestinal involvement, and 32% had proteinuria or hematuria.⁸⁵

Skin involvement is manifested as palpable purpura involving the lower extremities and, occasionally, the upper extremities, buttocks, and trunk. Biopsy specimens from skin lesions show leukocytoclastic vasculitis, and direct immunofluorescence testing may reveal deposits of IgA, and sometimes C3 and IgG within vessel walls. The skin findings generally fade without treatment. Ankles and knees tend to be more painful than other joints. Pain, nausea, vomiting, melena, and hematochezia herald gastrointestinal involvement, especially intussusception.

Kidney involvement may manifest only as microscopic hematuria and proteinuria, although some patients develop nephrotic syndrome. Renal involvement can be the most severe aspect of Henoch–Schönlein purpura.⁸⁴ In children, renal function generally returns to normal with treatment,⁸⁶ but in a cohort of 250 adults, 11% developed ESRD and another 27% developed chronic moderate to severe renal insufficiency.⁸⁵ A retrospective study of patients with Henoch–Schönlein purpura has shown increased renal morbidity to be associated with nephrotic syndrome, decreased factor XIII levels, hypertension, and certain biopsy findings (e.g., crescents, mesangial macrophages, tubulointerstitial fibrosis). Kidney biopsy specimens generally show glomerulonephritis, often with IgA deposits in the mesangium and vessel walls.⁸⁷ The association of IgM deposition in the skin with renal involvement in adults is unclear.^{88,89}

The pathogenesis of Henoch–Schönlein purpura is unknown. In one series, 62% of affected children had increased serum IgA levels.⁸⁴ Abnormal antigenic stimulation, possibly due to an infectious agent, may be a factor.⁸⁴ Elevated levels of vascular endothelial growth factor and endothelin have been reported,^{90,91} as have decreased levels of factor XIII.⁹² Abnormal cytokine levels (elevated tumor necrosis factor- α and interleukin-1 β) have been described.⁹³ High titers of IgA anti-endothelial cell antibodies are seen in patients with more severe renal involvement.⁹⁴

Most patients do not need treatment because the disease is self-limited. Joint pain may be ameliorated with analgesics. Severe gastrointestinal pain may benefit from systemic steroid treatment. Plasmapheresis and administration of steroids, intravenous immunoglobulin, and other immunosuppressants may decrease morbidity from renal involvement.^{83,84}

GENODERMATOSES

Several genodermatoses have prominent skin and renal findings (see Table 58.1). In Birt-Hogg-Dubé syndrome,^{95,96} patients have numerous flesh-colored to slightly tan, smooth papules on the face (Fig. 58.23) that histopathologically are fibrofolliculomas or trichodiscomas. Skin tag-like lesions may also be seen in the axillary areas.⁹⁷ Angiofibromas and perifollicular fibromas are other potential cutaneous lesions.⁹⁸ This autosomal dominant disorder is associated with renal cell carcinoma (e.g., chromophobe, hybrid, oncocytic and, rarely, clear cell types) and cysts, with a defect in folliculin, a product

Table 58.1 Genodermatoses With Associated Skin and Kidney Tumors

Syndrome	Skin Findings	Renal Findings
Birt-Hogg-Dubé syndrome	Fibrofolliculoma Trichodiscoma Skin tag-like lesion	Renal cell carcinoma Cysts
Tuberous sclerosis	Adenoma sebaceum	Angiomyolipoma Cysts
Von Hippel-Lindau syndrome	Port-wine stain	Clear cell renal cell carcinoma Cysts
Muir-Torre syndrome	Sebaceous neoplasia	Genitourinary carcinoma, including renal
Hereditary leiomyomatosis and renal cell carcinoma syndrome	Leiomyoma	Renal cell carcinoma



Fig. 58.23 Birt-Hogg-Dubé syndrome. Multiple whitish, smooth-surfaced papules (identified as fibrofolliculomas on biopsy) are present on the face. (Courtesy Yale Residents' Collection.)

of the *BHD* gene on chromosome 17. Folliculin likely functions as a tumor suppressor, and haploinsufficiency may be sufficient to lead to skin tumor formation.⁹⁹ Other manifestations include spontaneous pneumothorax.

In tuberous sclerosis,¹⁰⁰ another autosomal dominant disorder that is sometimes sporadic, there are also numerous facial papules clustered in the nasolabial areas and over the nose—so-called adenoma sebaceum or angiofibromas. Lesions are erythematous and smooth and may resemble acne. These lesions have histopathologic features similar to those of fibrous papules of the nose, with dilated vessels, stellate fibroblasts, and onion skin fibrosis around vessels and adnexal structures. Other skin findings include ash leaf-shaped hypopigmented macules and patches, small hypopigmented macules in a confetti-like pattern in the axillary areas, and periungual fibromas. Renal angiomyolipomas, cysts, or carcinoma may be seen. Patients may also have seizures or mental retardation. Genetic mutations are found in tuberin and hamartin, tumor suppressor proteins.

In von Hippel-Lindau syndrome, port-wine stains of the skin (seen in a minority of patients) are associated with ocular, cerebellar, medullary, and spinal hemangioblastomas, clear



Fig. 58.24 Multiple piloleiomyomas in a patient with Reed syndrome. (Courtesy Yale Residents' Collection.)

cell renal cell carcinoma and cysts, pheochromocytoma, pancreatic tumors and cysts, and testicular cysts. The disease has an autosomal dominant inheritance pattern, and ocular evaluation should begin at birth. Neurologic, otologic, and endocrine evaluation with abdominal ultrasonographic screening may be initiated at the age of 8 years or earlier if symptoms are present. The defective gene is *VHL*, a tumor suppressor gene on chromosome 3.¹⁰¹ The VHL protein regulates hypoxia-inducible factors 1 α and 2 α , and dysregulation leads to augmented angiogenesis in tumors.

In hereditary leiomyomatosis and renal cell carcinoma syndrome, cutaneous (Fig. 58.24) and uterine leiomyomas are associated with an aggressive renal cell carcinoma. Inheritance may be autosomal dominant, and mutations are in fumarate hydratase, a Krebs cycle enzyme that when defective, may lead to dysregulation of the same hypoxia-inducible factors as those in von Hippel–Lindau syndrome.¹⁰²

In Fabry disease,¹⁰³ numerous bright red, sometimes hyperkeratotic, papules are seen distributed in the bathing suit–trunk area. Such lesions may also be a manifestation of other genetic disorders, such as fucosidosis and other lipid storage disorders. Histopathologically, these lesions, termed “angiokeratomas,” are composed of dilated vessels that abut the undersurface of the epidermis. Other associated findings include paroxysmal pain, cornea verticillata, strokes, seizures, heart disorders, and chronic kidney failure. The defect is in α -galactosidase A, a product of the *GAL* gene on the X chromosome.

In Muir–Torre syndrome, multiple keratoacanthomas or a single sebaceous neoplasm (adenoma, carcinoma, or epithelioma) is associated with an internal malignancy. Usually, patients have a colorectal carcinoma, but carcinomas of the renal pelvis, ureter, and bladder are also associated. Muir–Torre syndrome is an autosomal dominant disorder with mutations described in *MSH2*, *MSH1* and, rarely *MSH6*.¹⁰⁴

Alport syndrome is the association of hematuric nephropathy, hearing defects, and ocular abnormalities. Although there are no skin findings, it is important to note that a simple skin biopsy can aid in the diagnosis. In Alport syndrome, there are mutations in the genes encoding type IV collagen. An absence of collagen IV may be seen in skin biopsy specimens from patients with X-linked Alport syndrome.¹⁰⁵

Box 58.4 Dermatologic Conditions That May Later Involve the Kidneys

Impetigo and streptococcal skin infection
Metastatic melanoma

DERMATOLOGIC CONDITIONS THAT MAY LATER INVOLVE THE KIDNEYS

See Box 58.4.

IMPETIGO AND STREPTOCOCCAL SKIN INFECTION

Although poststreptococcal glomerulonephritis is usually associated with a preceding streptococcal pharyngitis, streptococcal skin infections can be the initiating event. The renal symptoms may present 2 to 3 weeks or more after the skin infection, by which time the skin findings have generally resolved.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. A diagnosis of calciphylaxis can be supported by:
 - a. X-ray studies of soft tissues showing linear calcifications
 - b. Low serum calcium level
 - c. High protein C level
 - d. Leukocytoclastic vasculitis
 - e. Associated woody fibrosis of bilateral extremities

Answer: a

Rationale: Although a skin biopsy can be confirmatory of calciphylaxis, with calcium deposition within vessels, there can be a sampling issue. Also, biopsy can occasionally lead to poor wound healing. Radiologic studies can detect linear calcification of vessels and support the diagnosis.

2. A typical disorder associated with calcification within the skin is
 - a. Dermatomyositis
 - b. Diabetes mellitus
 - c. Intractable pruritus
 - d. Cryoglobulinemia
 - e. Porphyria cutanea tarda

Answer: a

Rationale: Calcification of the skin can be seen in renal failure; other situations that can present with calcinosis cutis include trauma (dystrophic calcification), dermatomyositis and, rarely, lupus erythematosus.

3. Typically, in porphyria cutanea tarda, blisters develop on the:

- a. Dorsal hands
- b. Soles
- c. Back
- d. Eyelid margin
- e. Oral mucosa

Answer: a

Rationale: Porphyria cutanea tarda can present with blisters, erosions, and/or crusts on the dorsal hands and forearms, sometimes with associated hypertrichosis of the face and hyperpigmentation of the skin.